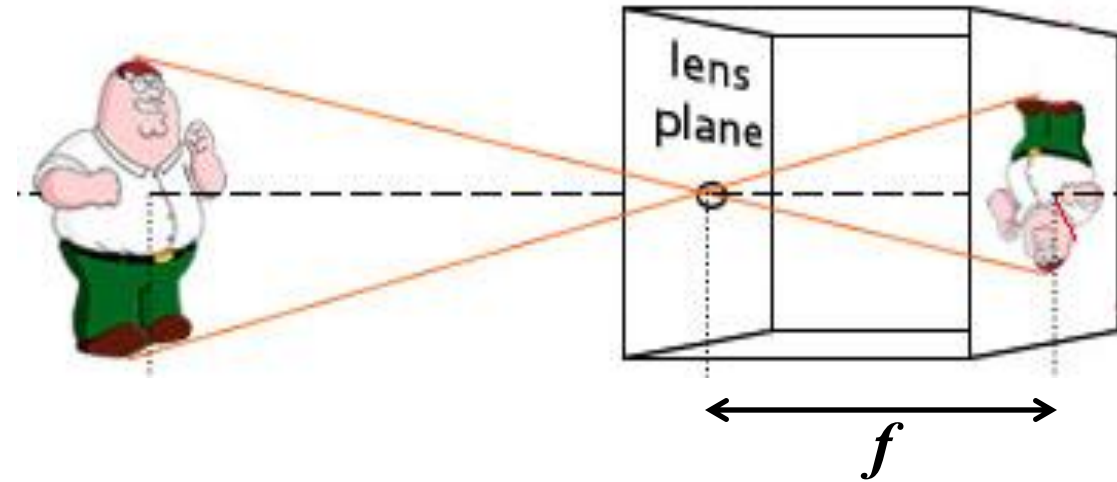


3D Imaging Geometry

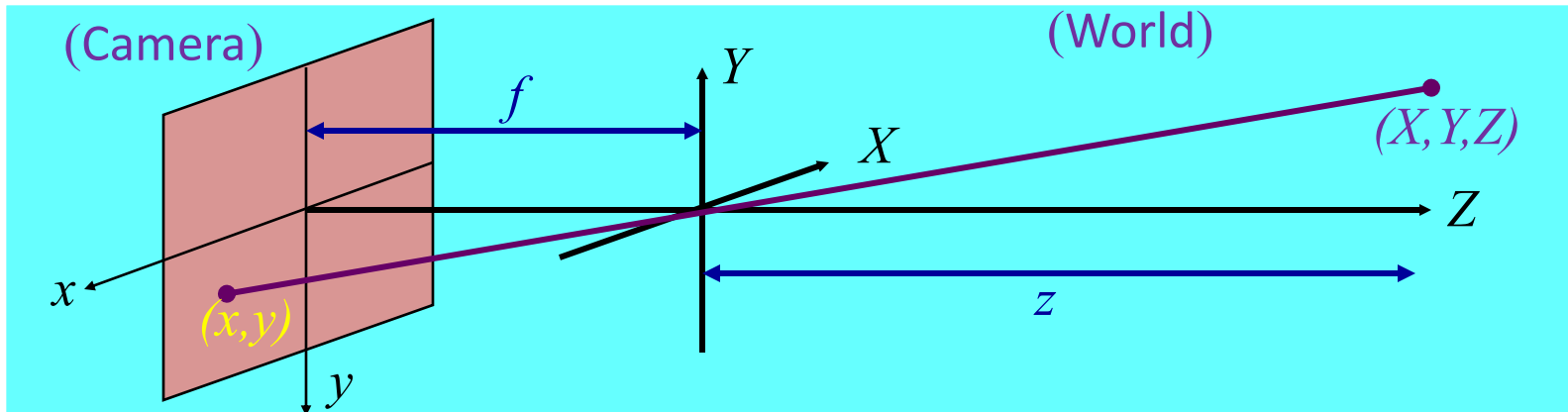
Pinhole Camera Model



- Pinhole model:
 - Captures pencil of rays – all rays through a single point
 - The point is called *Center of Projection* (COP), or *focal point*
 - A 2D image is formed on the Image Plane
 - Focal length f is distance from COP to Image Plane

Perspective Projection

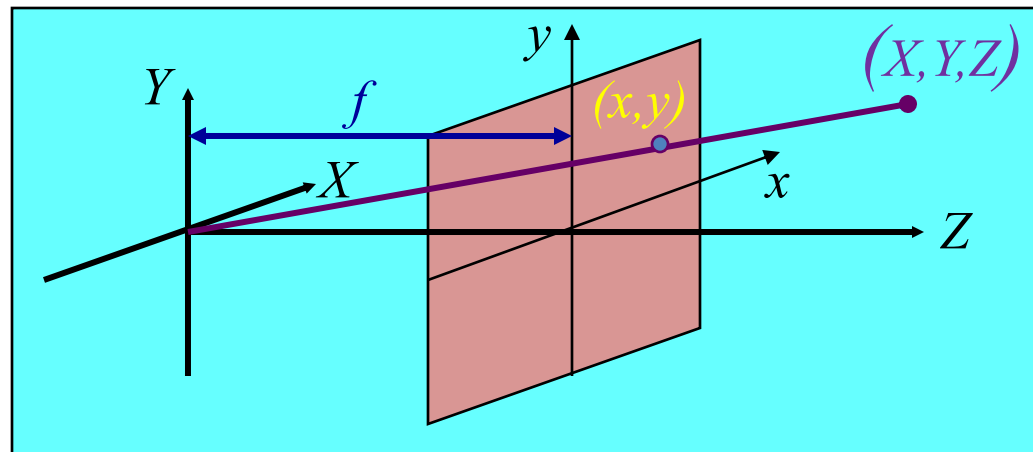
- Transforming the 3D world (X, Y, Z) into a 2D image (x, y)
 - Continuous Perspective Projection (optics)
 - All rays pass through one focal point (f = focal length)



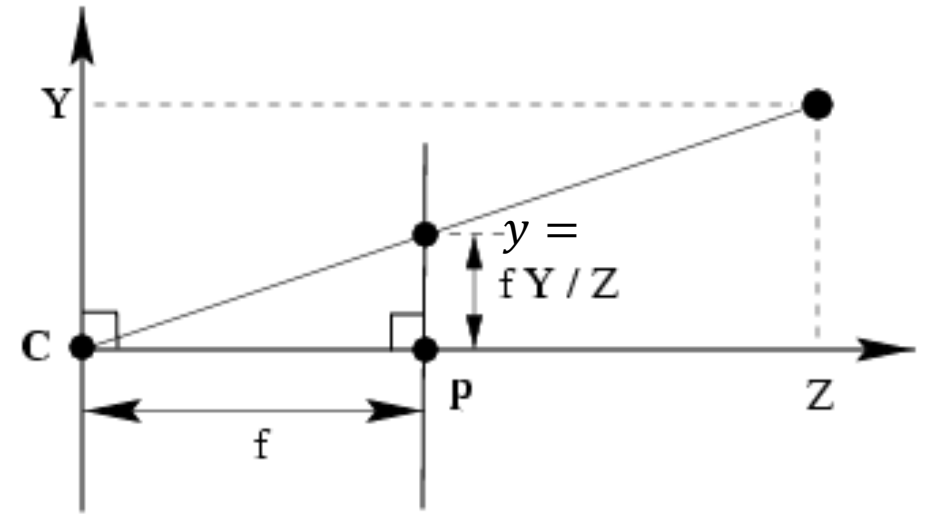
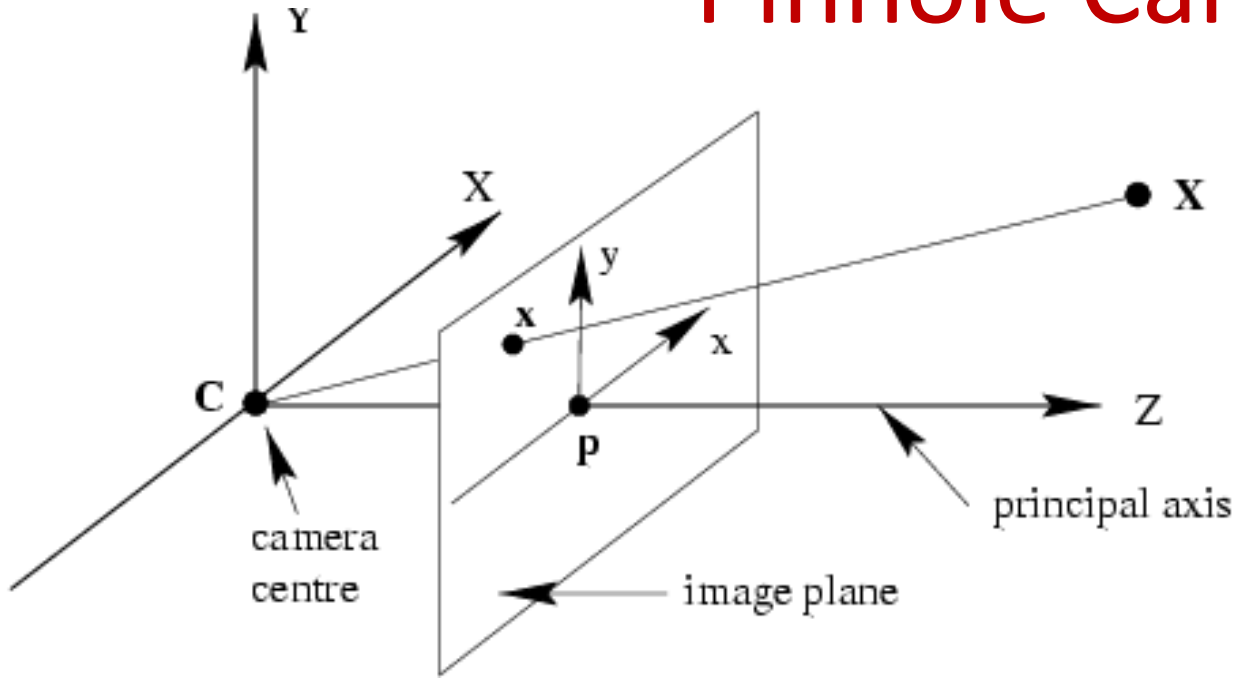
$$\frac{Y}{Z} = \frac{y}{f}$$



$$x = \frac{f}{Z} X$$
$$y = \frac{f}{Z} Y$$



Pinhole Camera Model



$$\mathbf{x}_3 = \mathbf{M}_{3 \times 4} \mathbf{X}_4$$

$$(X, Y, Z) \mapsto (x, y) = \left(f \frac{X}{Z}, f \frac{Y}{Z}\right)$$

$$\begin{pmatrix} fX/Z \\ fY/Z \end{pmatrix} \sim \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Homogenous coordinates:
camera matrix $\mathbf{M}$ is 3×4 ,
 projecting 3D world to 2D
 image

Camera Matrix & Homogenous Coordinates

When world and camera use same X, Y axes, and origin of world axes ($X=0, Y=0, Z=0$) is at camera's optical center:

Camera Matrix

$$\begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Homogenous Coordinates

	Regular	Homogenous
3D	$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$	$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$
2D	$\begin{pmatrix} f \frac{X}{Z} \\ f \frac{Y}{Z} \end{pmatrix}$	$\begin{pmatrix} fX \\ fY \\ Z \end{pmatrix}$

General Camera Matrix M

A camera in general position, not at world origin

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

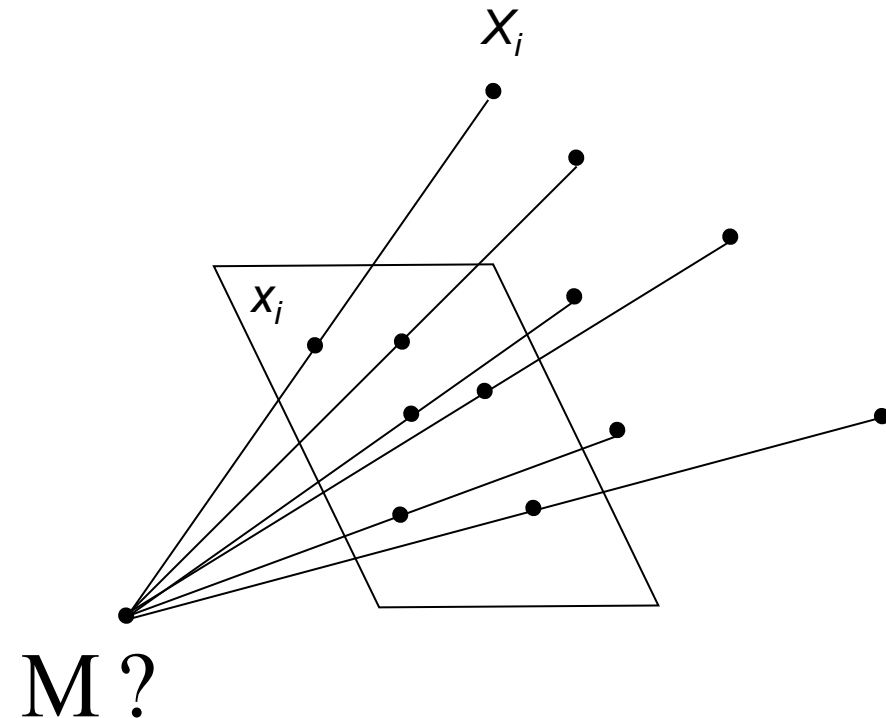
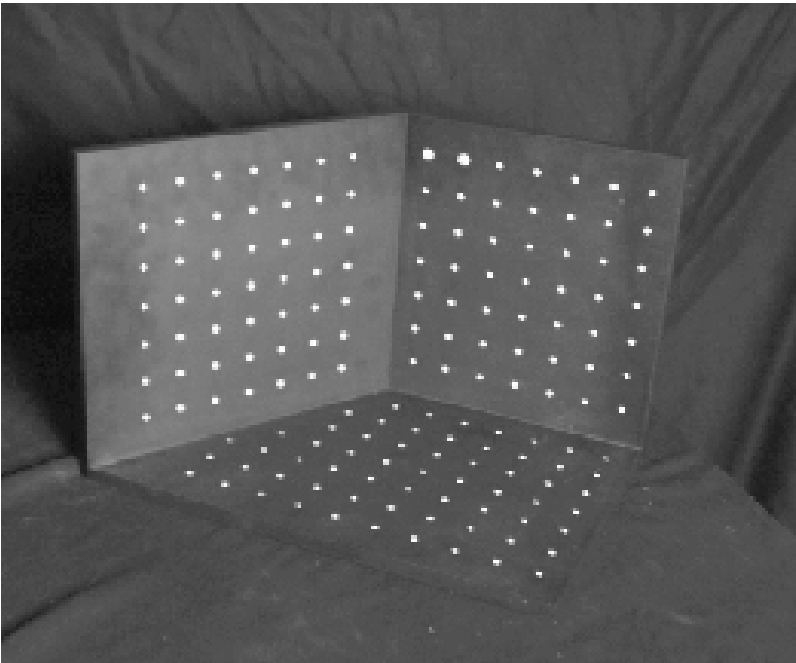
- Always 1 in lower right-hand corner
- 11 Degrees of Freedom
- $\sim$ is projective equality: dividing by 3rd parameter

Camera Matrix M

- In General, M has 11 degrees of freedom (12 parameters, but scale is arbitrary)
- 5 intrinsic (camera parameters): f_x, f_y , Center (c_x, c_y) , skew
- 6 extrinsic (camera location): 3 rotation, 3 translation
- One correspondence between a 2D image point (x, y) to a 3D world point (X, Y, Z) gives two independent linear equations (m_{ij} are unknowns), one for x and one for y .
- At least 6 correspondences between 3D points and image points needed to compute M

Camera Calibration

- Given $n \geq 6$ points with known 3D coordinates X_i and known image projections x_i , estimate the camera parameters by solving the equations



Camera Matrix Equations

- Camera maps each 3D points (X_i, Y_i, Z_i) to an image point (x_i, y_i)

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \sim \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$x_i = \frac{m_{00}X_i + m_{01}Y_i + m_{02}Z_i + m_{03}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + 1}$$

$$y_i = \frac{m_{10}X_i + m_{11}Y_i + m_{12}Z_i + m_{13}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + 1}$$

Camera Matrix Calibration

- Estimate the 11 unknowns in the $\mathbf{M}$ matrix using known 3D points (X_i, Y_i, Z_i) and corresponding image points (x_i, y_i)

$$x_i = \frac{m_{00}X_i + m_{01}Y_i + m_{02}Z_i + m_{03}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + 1}$$

$$y_i = \frac{m_{10}X_i + m_{11}Y_i + m_{12}Z_i + m_{13}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + 1}$$

- What are the unknowns?
- Why are these linear equations?

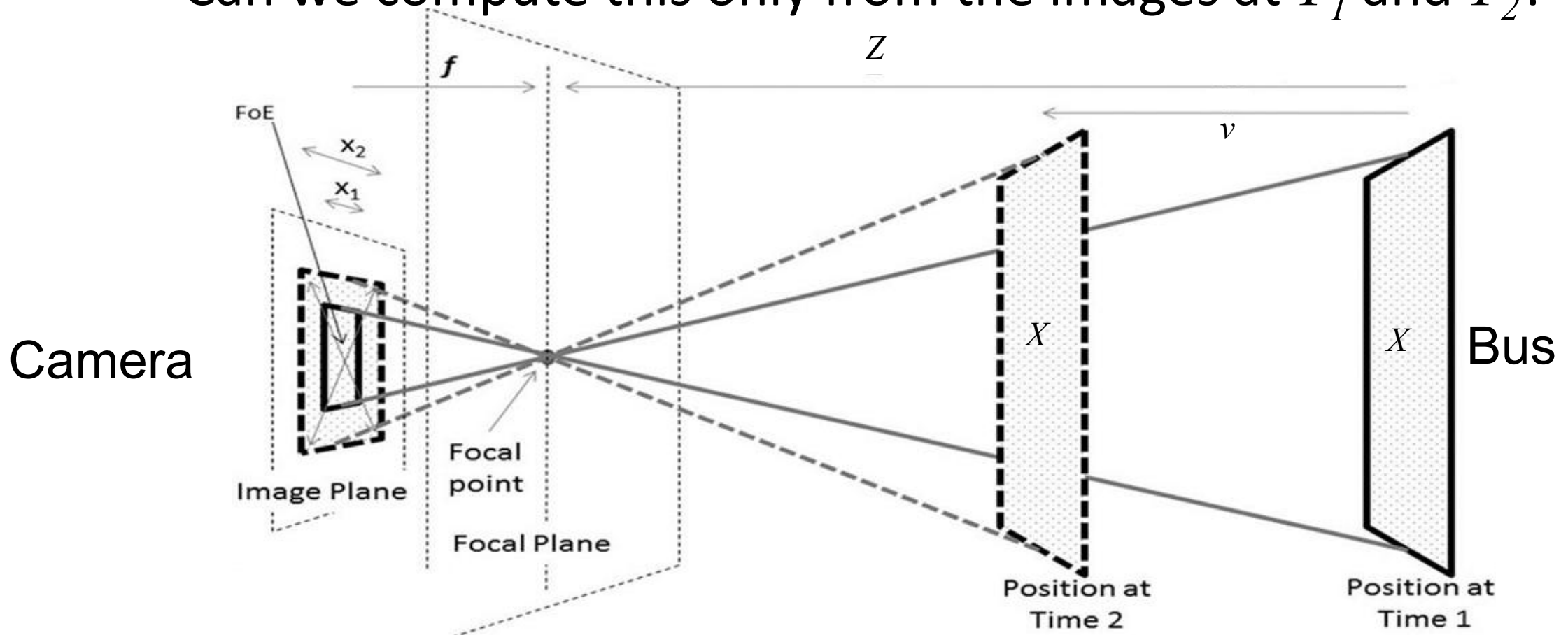
Time to Collision

- A bus is moving towards the camera. When will they collide?
Bus width doubled after 2 second.



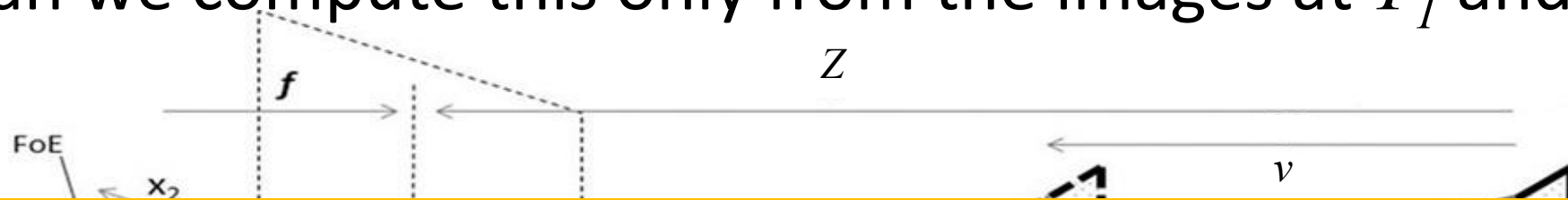
Time to Collision

- A bus of width X and distance Z is moving towards the camera with speed v . When will they collide?
- Can we compute this only from the images at T_1 and T_2 ?



Time to Collision

- A Bus of width X and distance Z is moving towards the camera with speed v . When will they collide?
- Can we compute this only from the images at T_1 and T_2 ?



- We do not know the size of the bus
- We do not know the speed of the bus
- We do not know the distance of the bus
- We do not know the camera matrix (f)

Camera

Bus

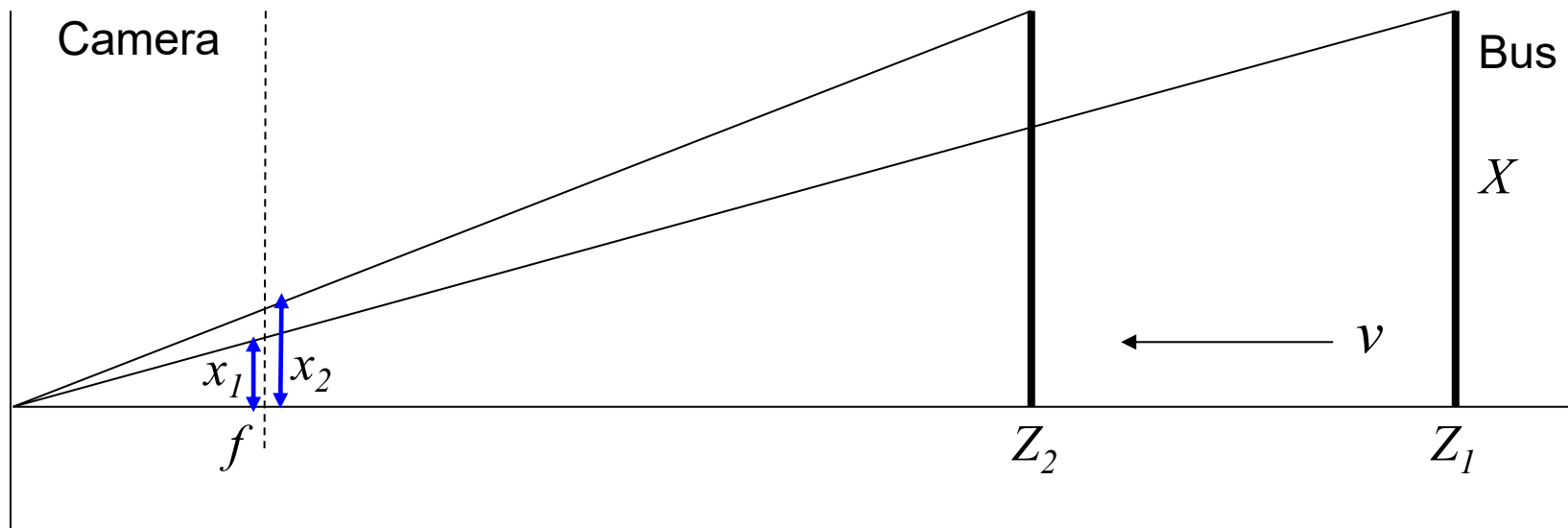


Time to Collision (TTC)

- At frame 1 the image width of bus is x_1
- At frame 2 the image width of bus is x_2

$$\text{TTC} = \frac{Z_1}{Z_1 - Z_2} \underset{\substack{\uparrow \\ \text{Divide by } Z_1 Z_2}}{=} \frac{1/Z_2}{1/Z_2 - 1/Z_1} \underset{\substack{\uparrow \\ \text{Multiply by } fX}}{=} \frac{fX/Z_2}{fX/Z_2 - fX/Z_1} \underset{\substack{\uparrow \\ \text{Projection}}}{=} \frac{x_2}{x_2 - x_1} \text{ (time is \#frames)}$$

Examples: ♦ $x_2 = x_1 \rightarrow \text{TTC} = \infty$ ♦ $x_2 = 2x_1 \rightarrow \text{TTC} = 2$



Epipolar Geometry: Two Cameras & Three Questions:

(i) Correspondence Geometry:

Given an image point x in the first image, how does this constrain the position of the corresponding point x' in the second image?

(ii) Camera Geometry (camera motion):

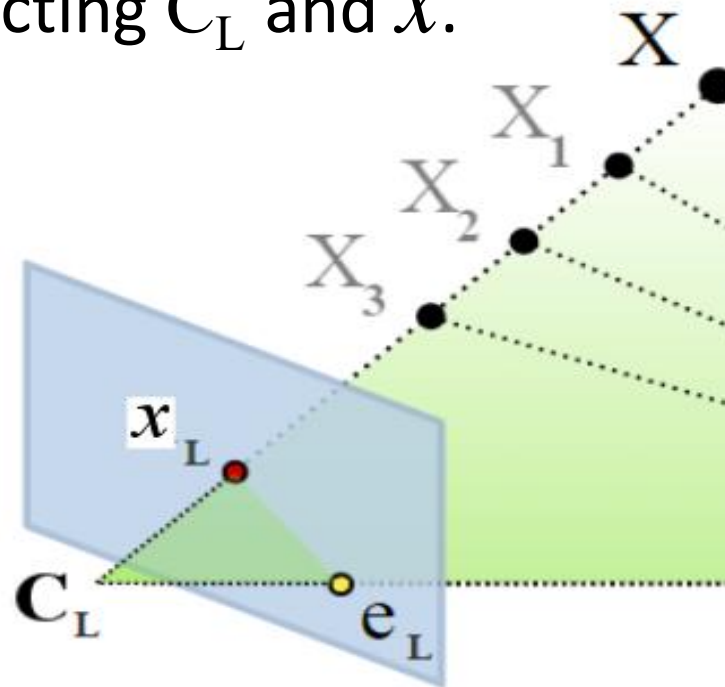
Given a set of corresponding image points between images, $\{x_i \leftrightarrow x'_i\}$, $i=1, \dots, n$, what are the cameras matrices M and M' for the two views?
Or: what is the geometric transformation between the views?

(iii) Scene Geometry (structure, stereo):

Given corresponding image points between images, $x_i \leftrightarrow x'_i$ and cameras M, M' , what is the position of the 3D point X in world coordinates?

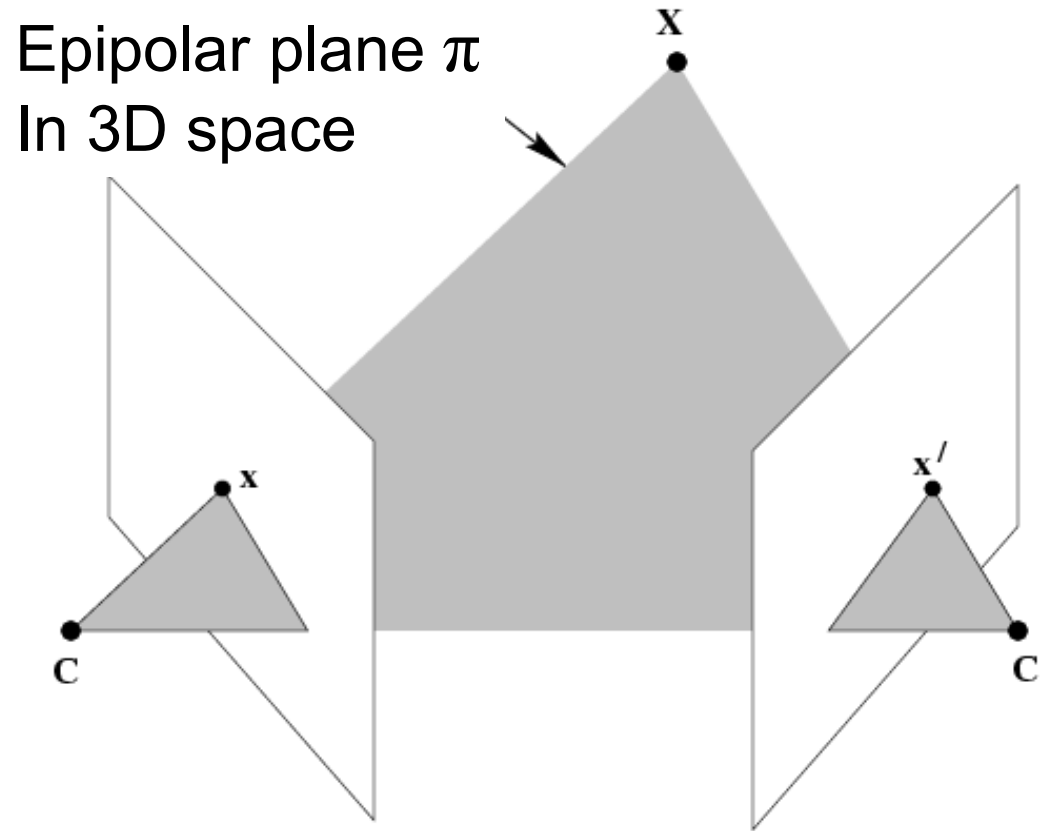
Epipolar Line for Image Point x

- Given image point x , it can originate from any 3D point X along a straight line through C_L and x . (C_L = center of left camera)
- Given image point x , the corresponding point x' in the second image can be anywhere along the epipolar line l' , which is the projection of the line connecting C_L and x .



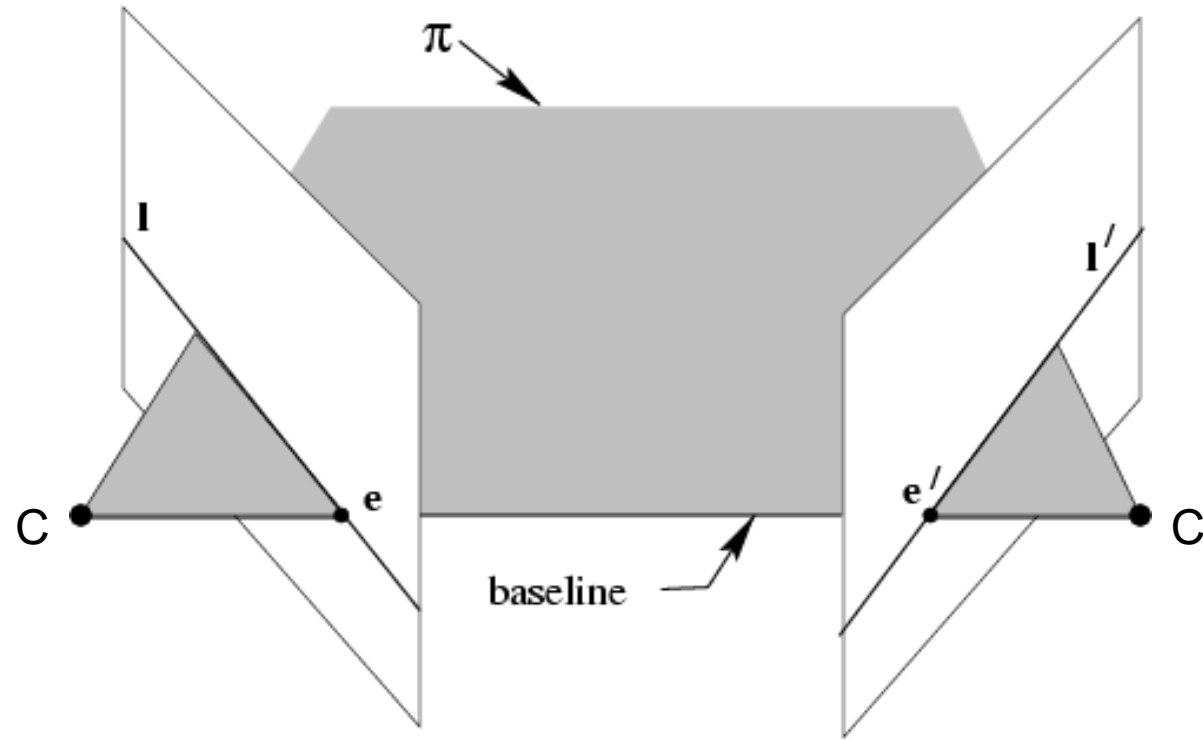
Left view

The Epipolar Plane



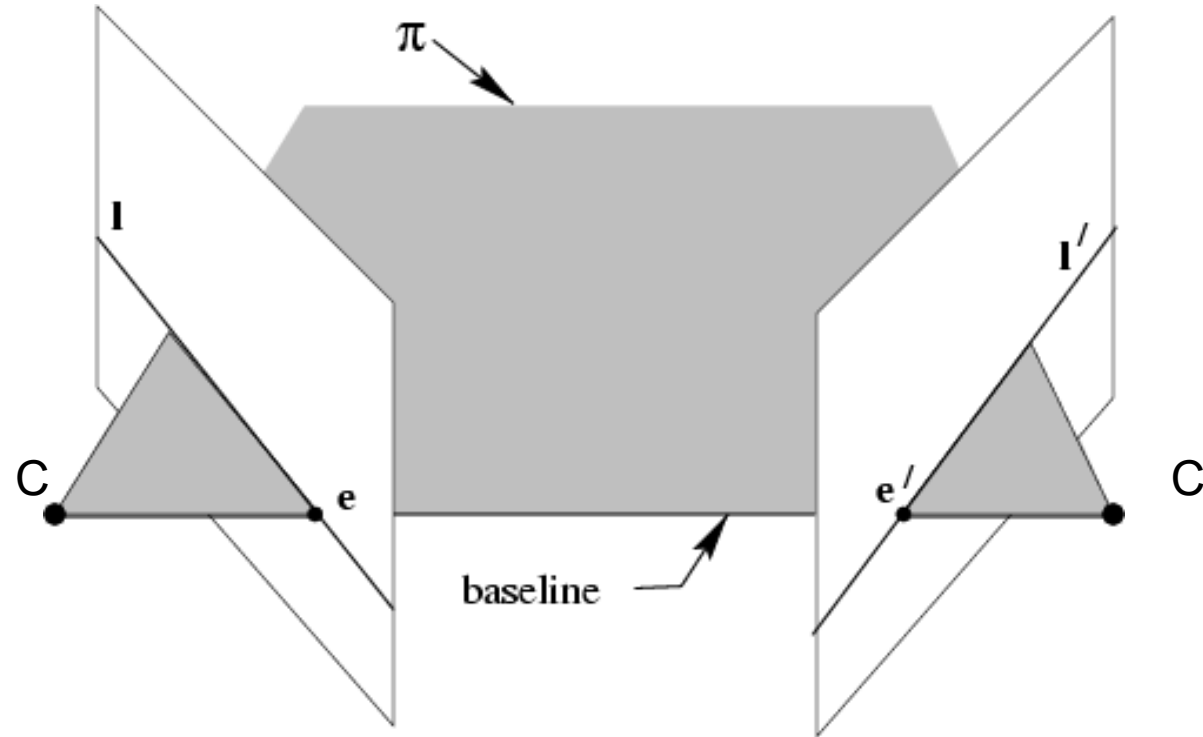
- Two cameras with centers C , C' view a 3D point X in image points x , x' . All these points are on the epipolar plane π .
- The epipolar plane π is determined by the points: C , C' and X

The Epipolar Geometry



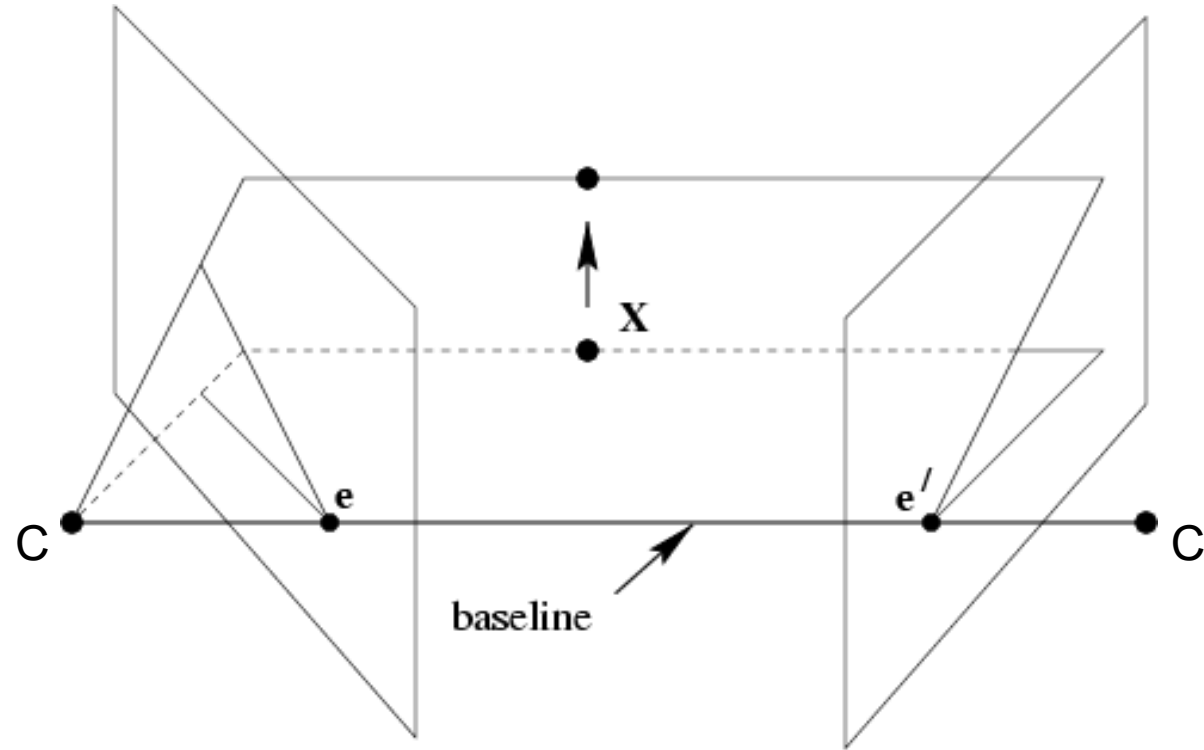
- The camera baseline, connecting the two camera centers, intersects the image planes at the epipoles e and e' .
- Any plane π containing the baseline is an epipolar plane.
- The epipole is the image of the center of the other camera

The Epipolar Geometry



- All points on the epipolar plane π project to the epipolar lines l and l'
- Can the epipolar plane be a physical plane, e.g., the floor?

The Epipolar Geometry



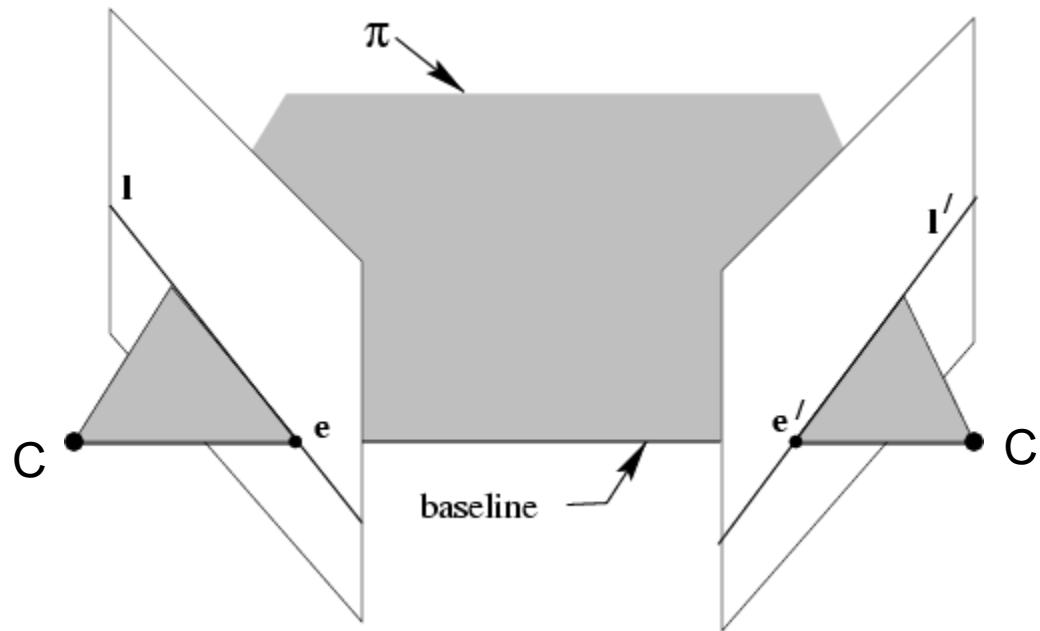
- When point X is in other locations, we get a family of epipolar planes π and epipolar lines l and l' .
- All epipolar planes (and lines) intersect the epipoles e and e' .

The Epipolar Geometry

Epipoles e, e'

= intersection of the baseline with an image plane

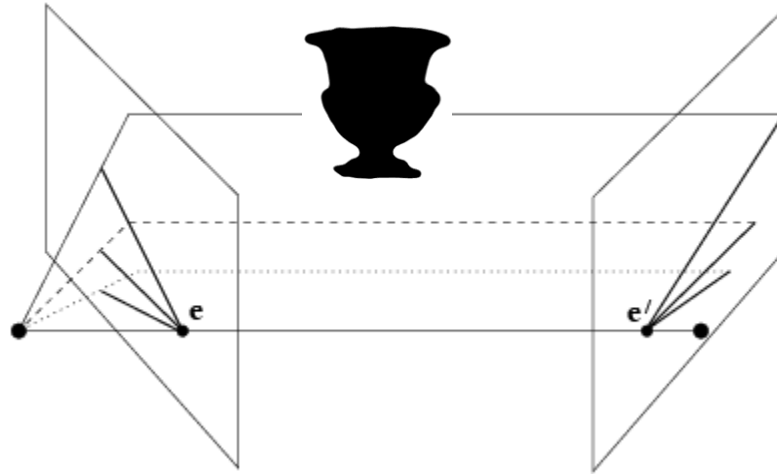
= The image of the projection center (pinhole) of the other camera



Epipolar plane = a plane containing the baseline (1-D family)

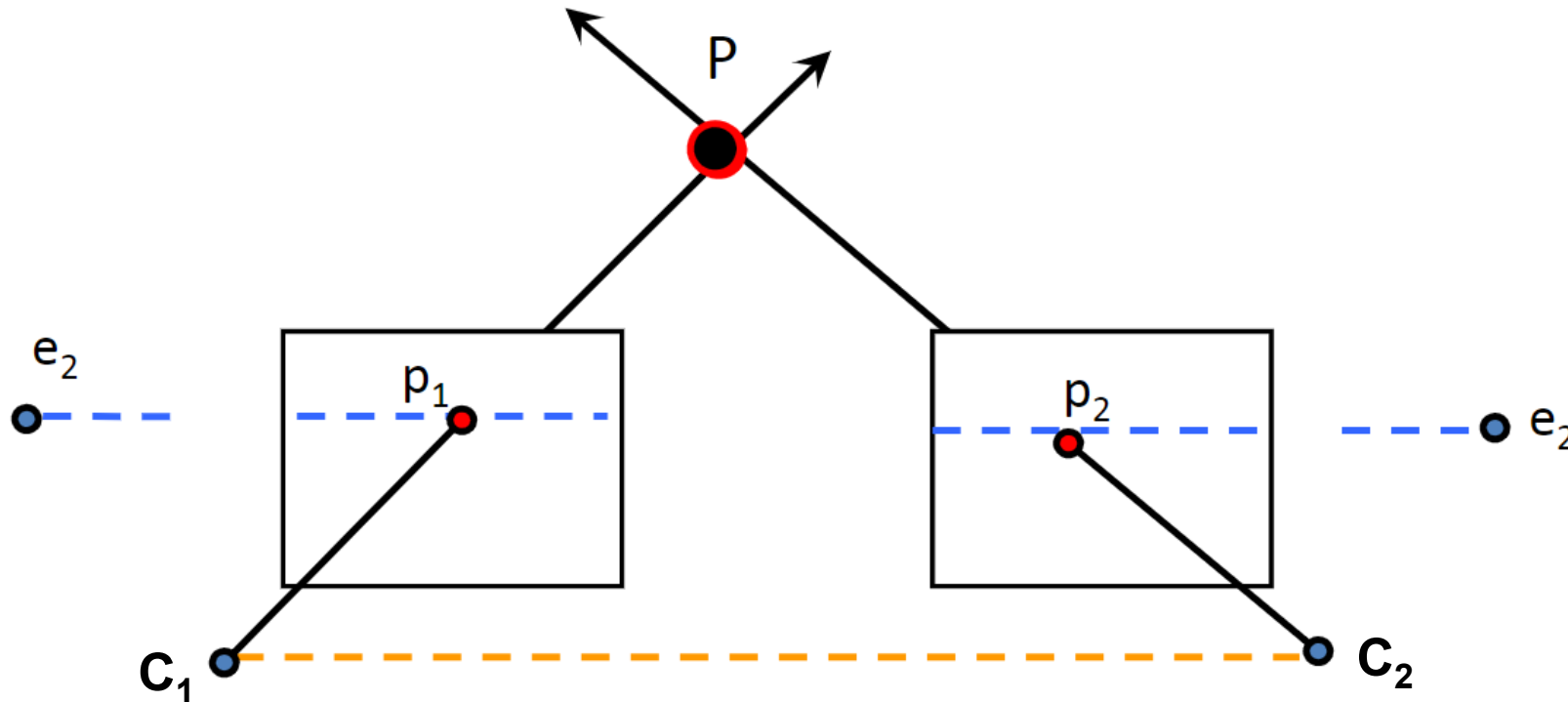
Epipolar line = intersection of an epipolar plane with the image plane
(always come in corresponding pairs)

Example: Converging Cameras



Which one of the two cameras is on the left? On the right?

Camera Motion Parallel to Image Plane



- The baseline intersects the image planes at infinity
- Epipoles are at infinity
- Epipolar lines parallel to x axis

Epipolar Lines – Parallel Translation

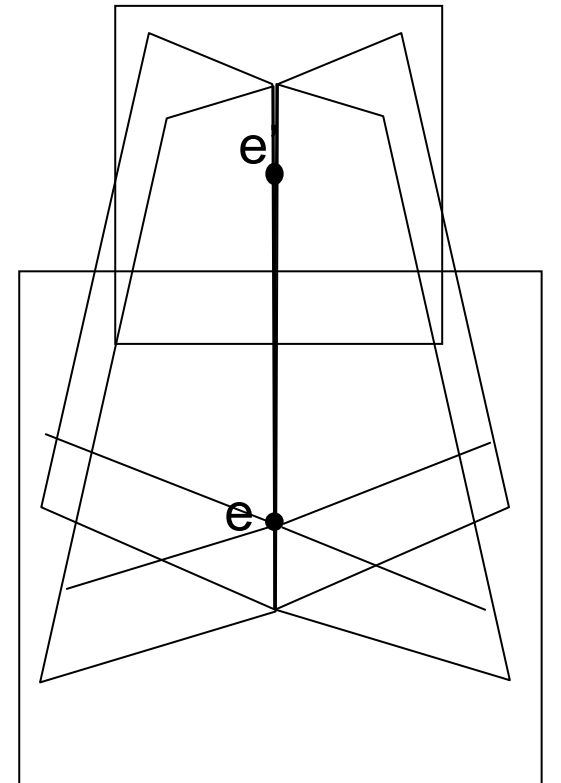
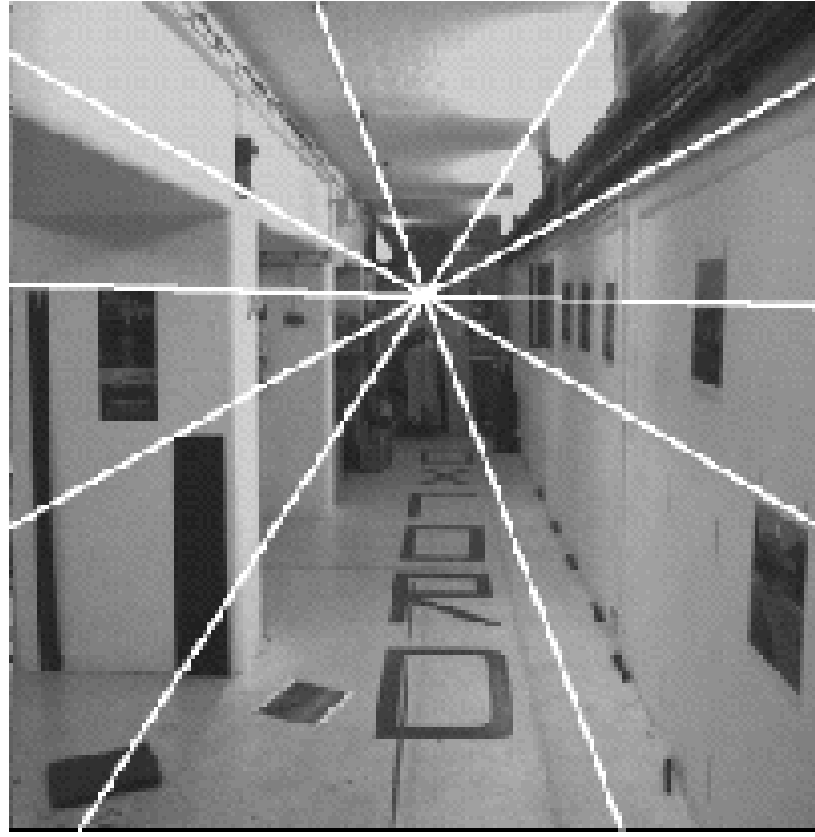
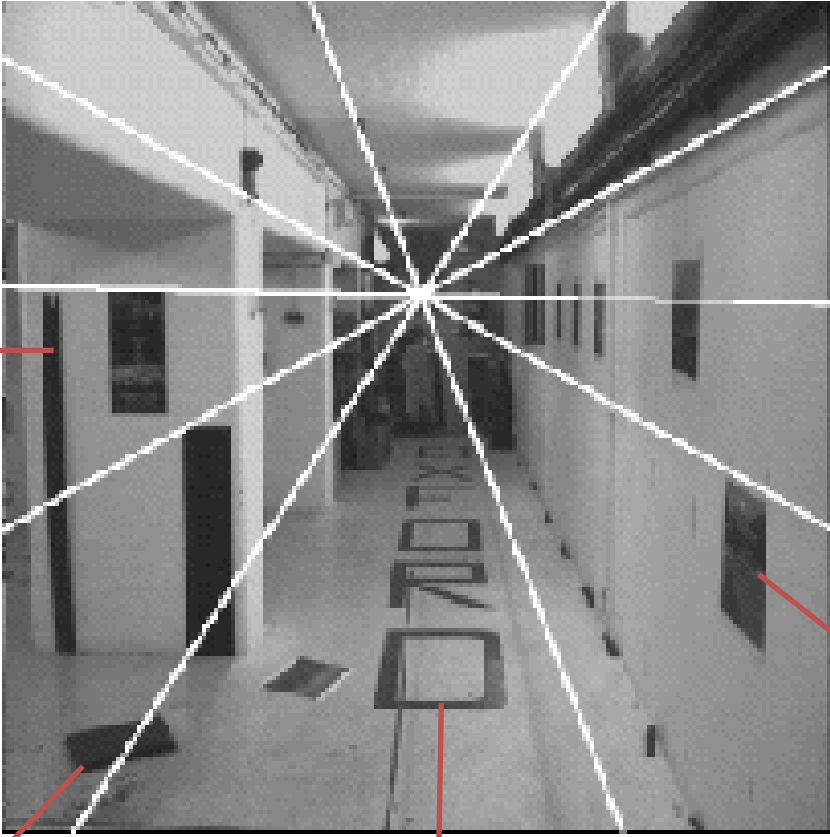


- Epipolar lines are Parallel in direction of translation
- Two corresponding epipolar lines go through same scene points

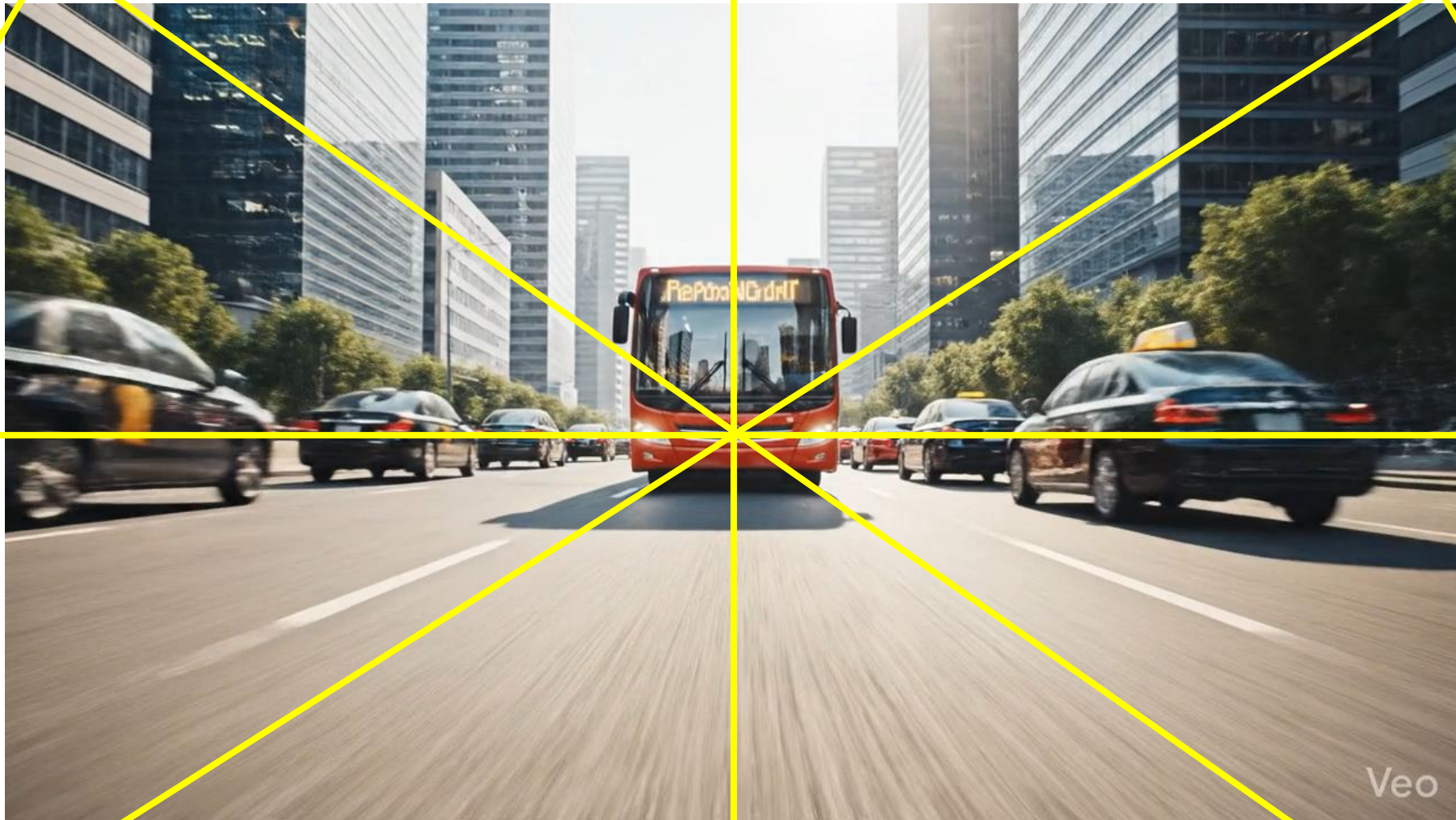
Example: Forward Camera Motion

1st frame

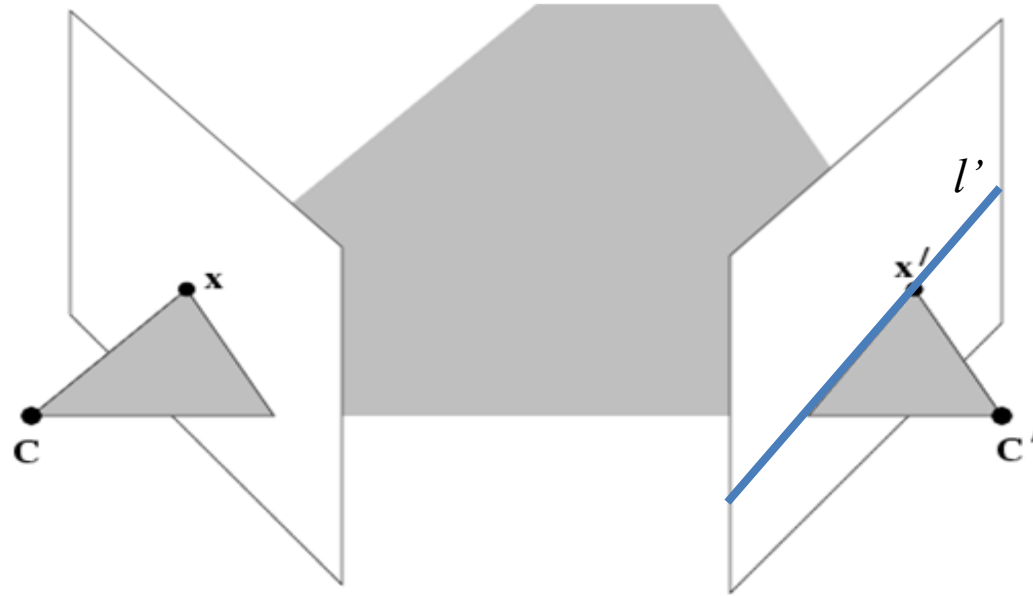
2nd frame



Example: Forward Camera Motion



Fundamental Matrix



For any two cameras viewing the same scene, there exists a 3×3 “Fundamental Matrix” F , such that for any two corresponding image points x and x' :

$$x'^T F x = 0$$

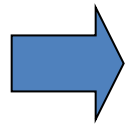
Given x and F , all points on the epipolar line l' satisfy the above equation.

Compute Fundamental Matrix

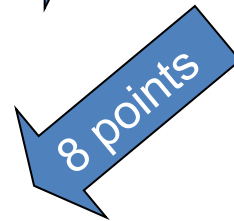
The Eight-point Algorithm

One point correspondence: $\mathbf{x} = (u, v, 1)^T$, $\mathbf{x}' = (u', v', 1)^T$

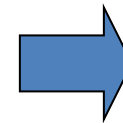
$$(u, v, 1) \begin{pmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{pmatrix} \begin{pmatrix} u' \\ v' \\ 1 \end{pmatrix} = 0$$



$$(uu', uv', u, vu', vv', v, u', v', 1) \begin{pmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{pmatrix} = 0$$



$$\begin{pmatrix} u_1u'_1 & u_1v'_1 & u_1 & v_1u'_1 & v_1v'_1 & v_1 & u'_1 & v'_1 \\ u_2u'_2 & u_2v'_2 & u_2 & v_2u'_2 & v_2v'_2 & v_2 & u'_2 & v'_2 \\ u_3u'_3 & u_3v'_3 & u_3 & v_3u'_3 & v_3v'_3 & v_3 & u'_3 & v'_3 \\ u_4u'_4 & u_4v'_4 & u_4 & v_4u'_4 & v_4v'_4 & v_4 & u'_4 & v'_4 \\ u_5u'_5 & u_5v'_5 & u_5 & v_5u'_5 & v_5v'_5 & v_5 & u'_5 & v'_5 \\ u_6u'_6 & u_6v'_6 & u_6 & v_6u'_6 & v_6v'_6 & v_6 & u'_6 & v'_6 \\ u_7u'_7 & u_7v'_7 & u_7 & v_7u'_7 & v_7v'_7 & v_7 & u'_7 & v'_7 \\ u_8u'_8 & u_8v'_8 & u_8 & v_8u'_8 & v_8v'_8 & v_8 & u'_8 & v'_8 \end{pmatrix} \begin{pmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \end{pmatrix} = - \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$



Minimize:

$$\sum_{i=1}^N (x_i^T F x'_i)^2$$

under the constraint

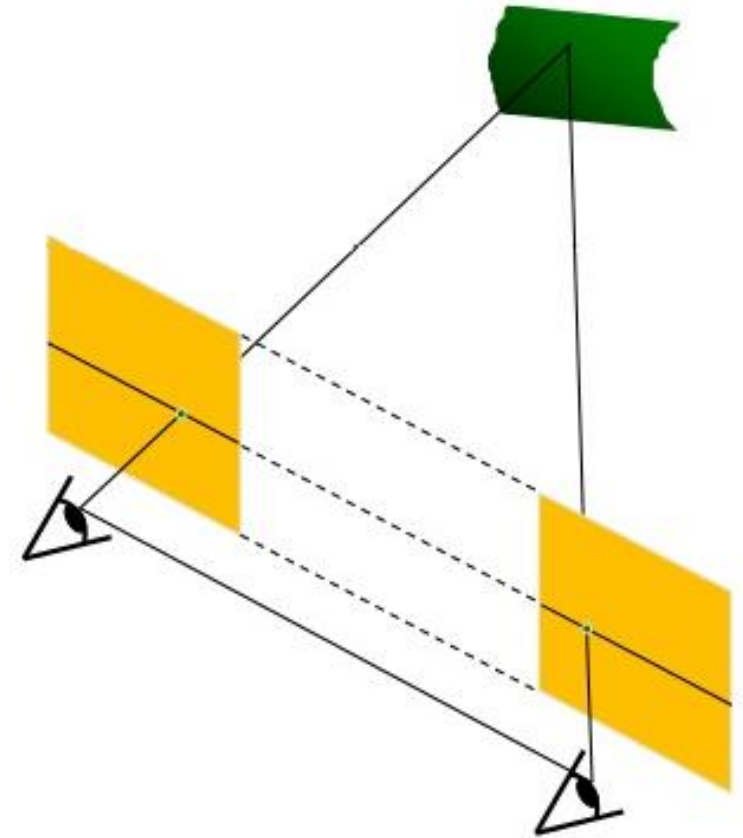
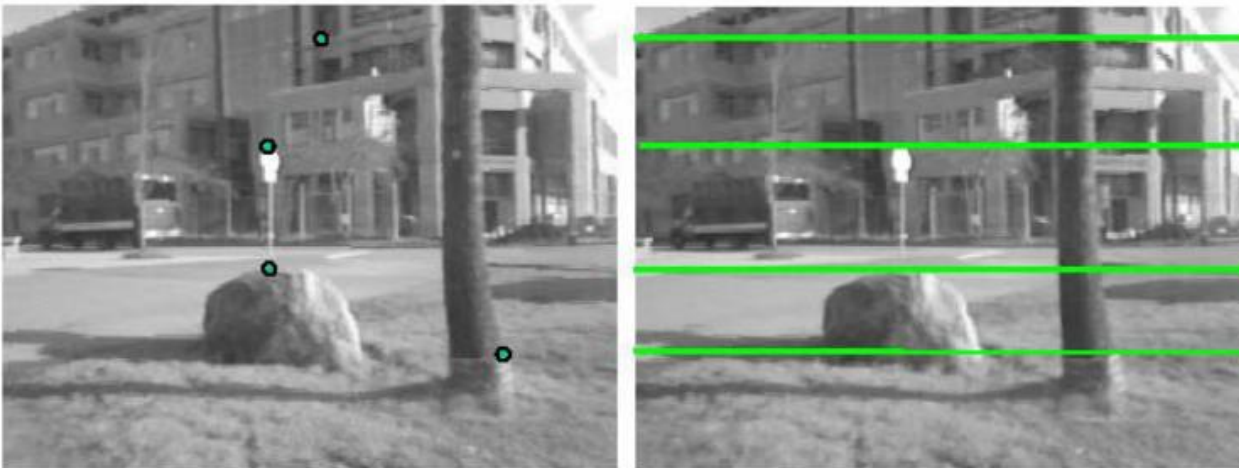
$$F_{33} = 1$$

3D Stereo Reconstruction

- Assume we know the viewing geometry (extrinsic parameters) and the intrinsic parameters (Calibrated camera)
- Find point correspondences exploiting epipolar geometry
 - Search for correspondences only on epipolar lines
- Compute depth by triangulation

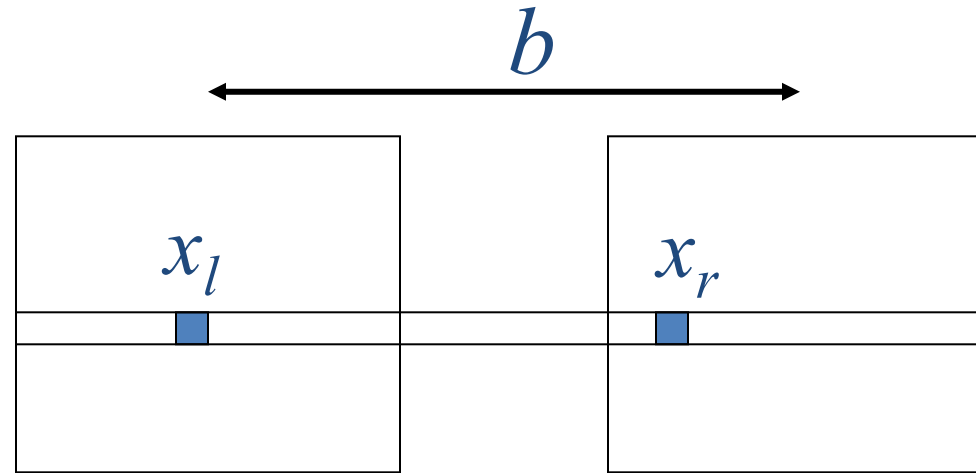
Depth from Stereo

- Two Cameras
- Same image plane
- Parallel optical axis
- Horizontal epipolar lines



Depth from Stereo

- 3D point



Point projected to left image

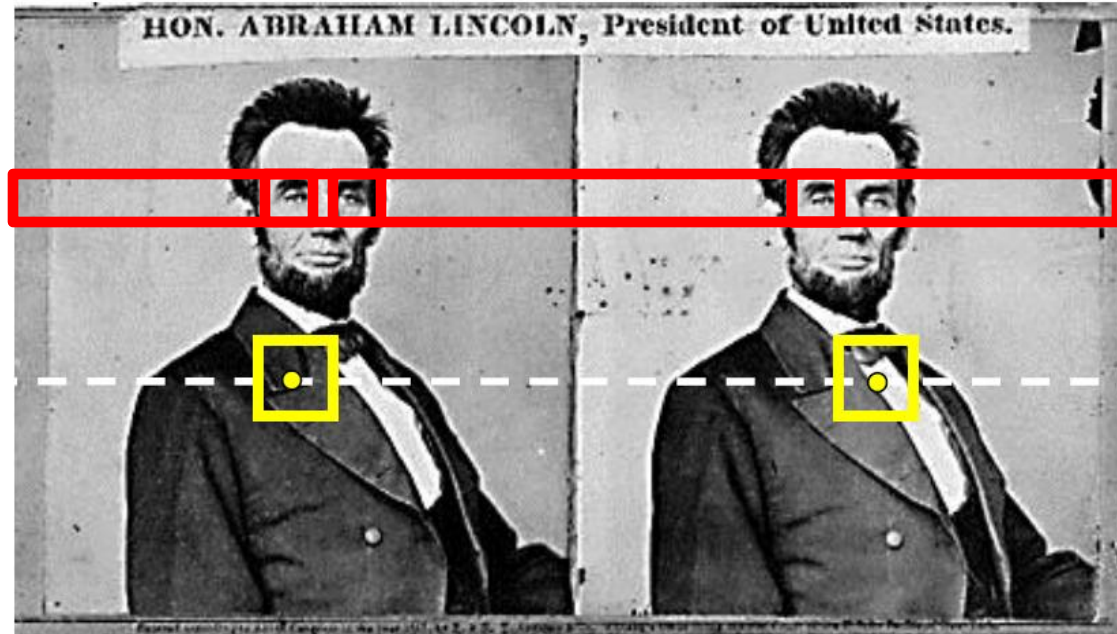
Point projected to right image

Baseline b : The distance between camera centers (Usually pure X translation)

Disparity d : the difference in image location of the same 3D point when viewed by two different cameras.

$$d = x_l - x_r$$

Basic Stereo Algorithm

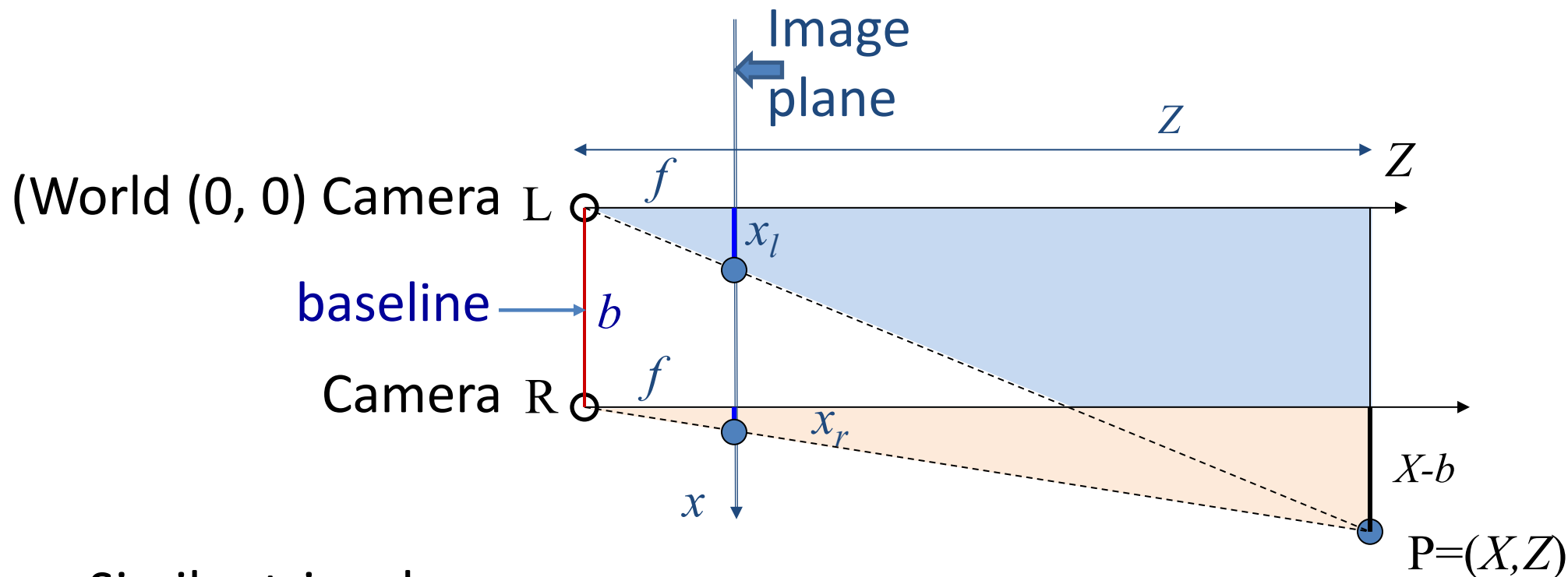


For each epipolar line

For each pixel in the left image – take a window around this pixel

- Compare this window with every pixel on the epipolar line in the right image
- Pick the pixel in the right image having window with highest correlation
- Use Ordering Constraint (left-right relationships are mostly preserved)

Depth from Stereo: Parallel Optic Axes



Similar triangles:

$$\frac{X}{Z} = \frac{x_l}{f}$$

$$\frac{X-b}{Z} = \frac{x_r}{f}$$

$$\Rightarrow Z = \frac{fb}{x_l - x_r} = \frac{fb}{d} \quad X = Z \frac{x_l}{f}$$

Parallax: $d = x_l - x_r = \frac{fb}{Z}$

Block Matching $\approx$ Optical Flow

How do we determine correspondences?

- *block matching* or *SSD* (sum squared differences)

$$E(x, y; d) = \sum_{(x', y') \in N(x, y)} [I_L(x' + d, y') - I_R(x', y')]^2$$

d is the *disparity* (horizontal motion)



How big should the neighborhood be?

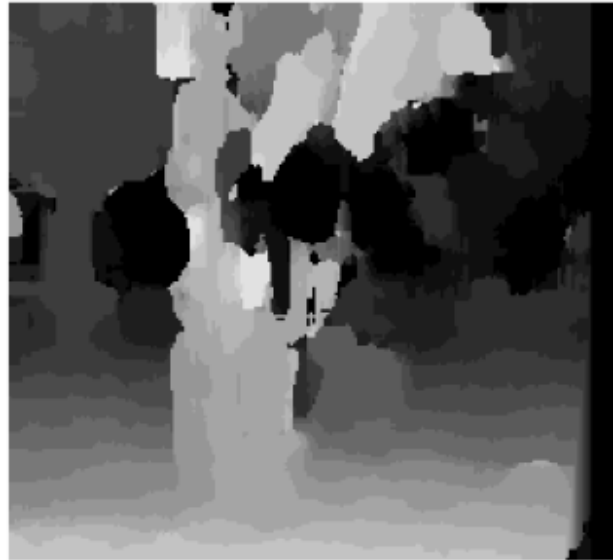
Neighborhood Size for Matching

Smaller neighborhood: more details

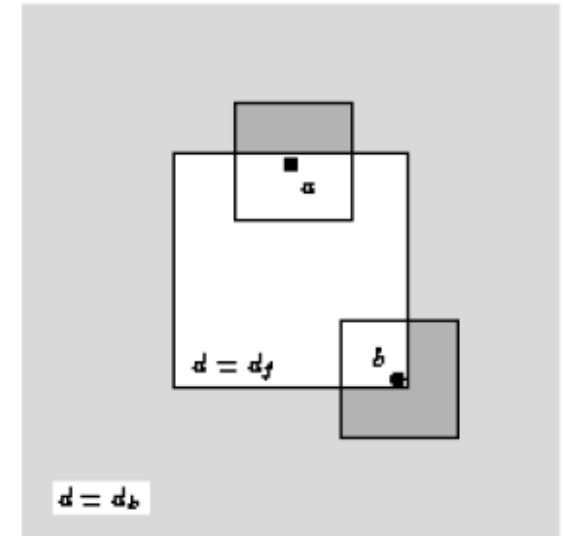
Larger neighborhood: fewer isolated mistakes



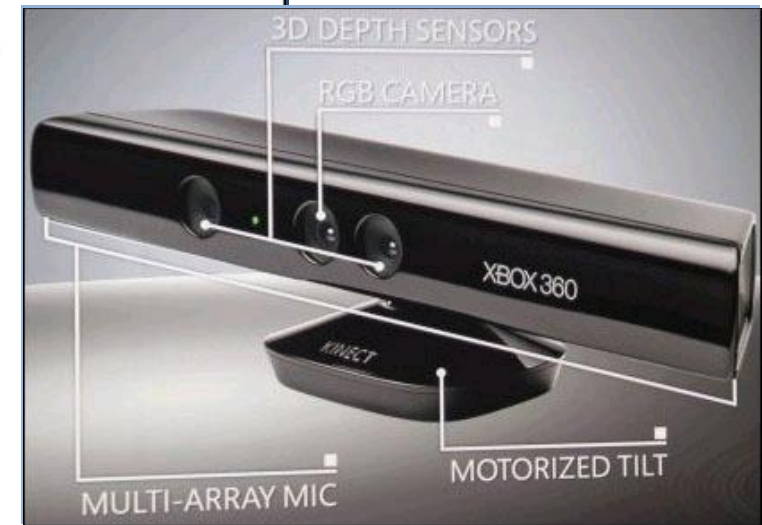
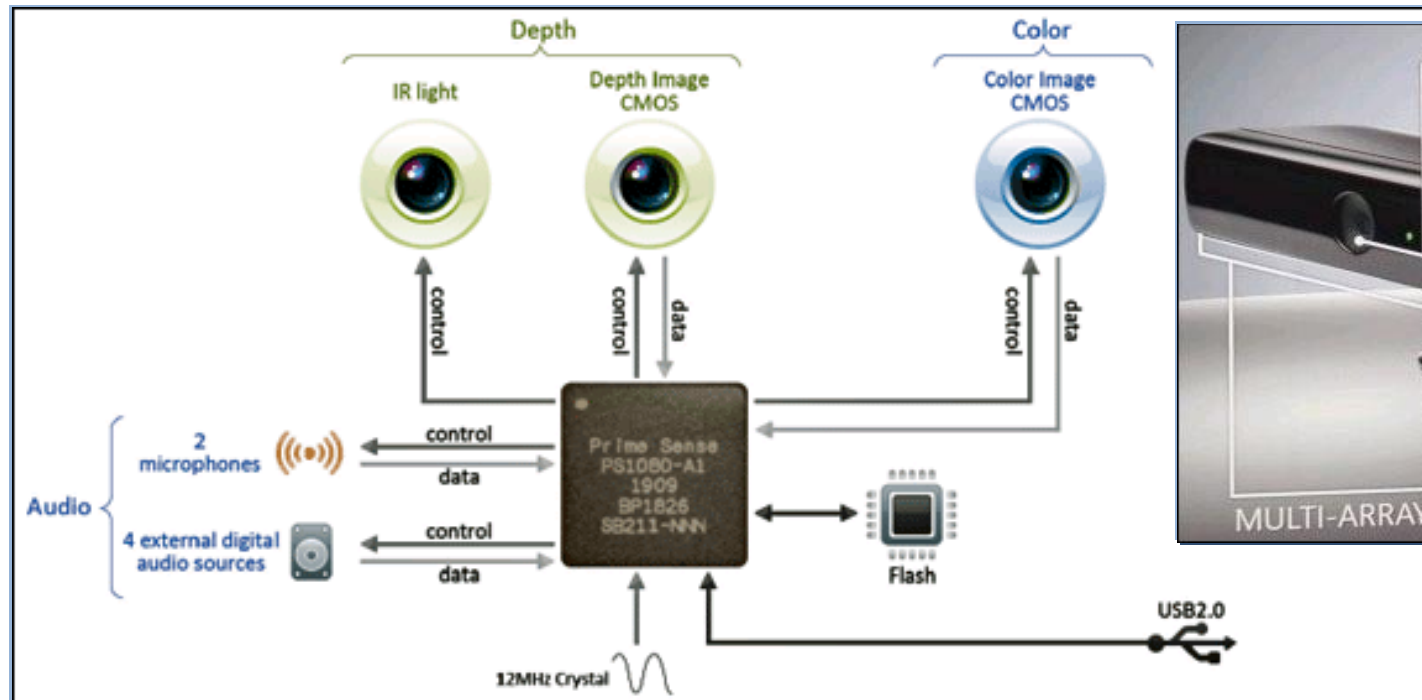
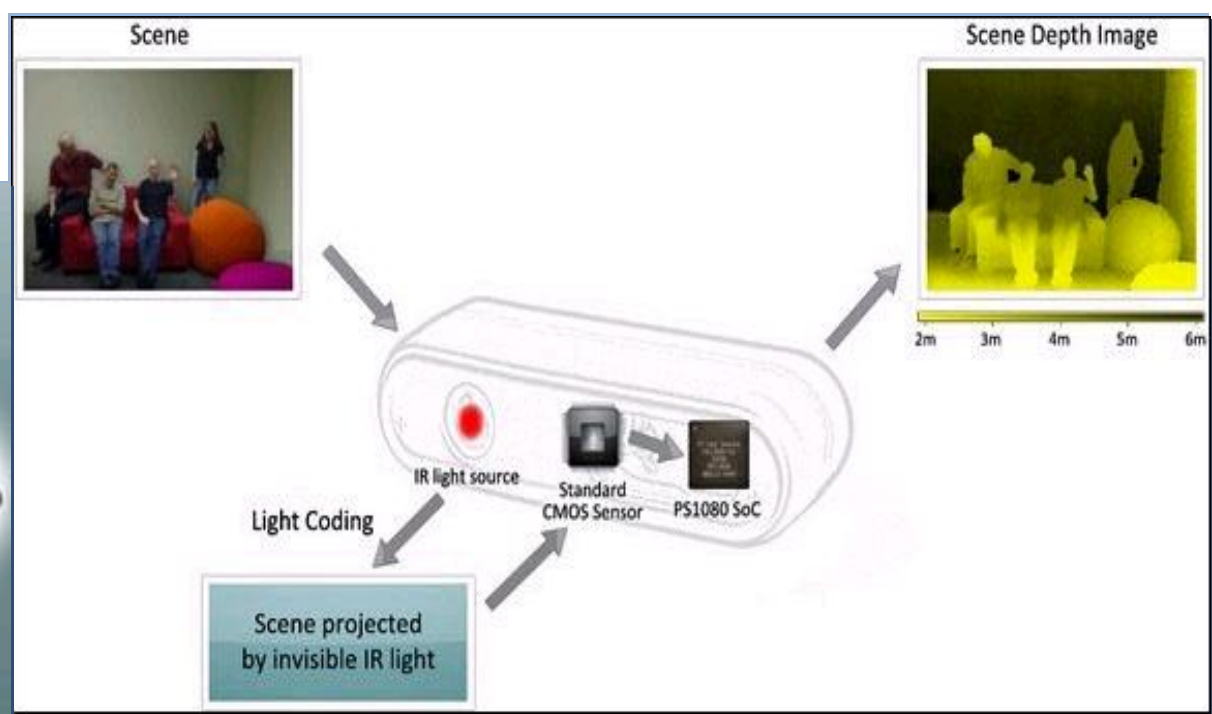
$w = 3$



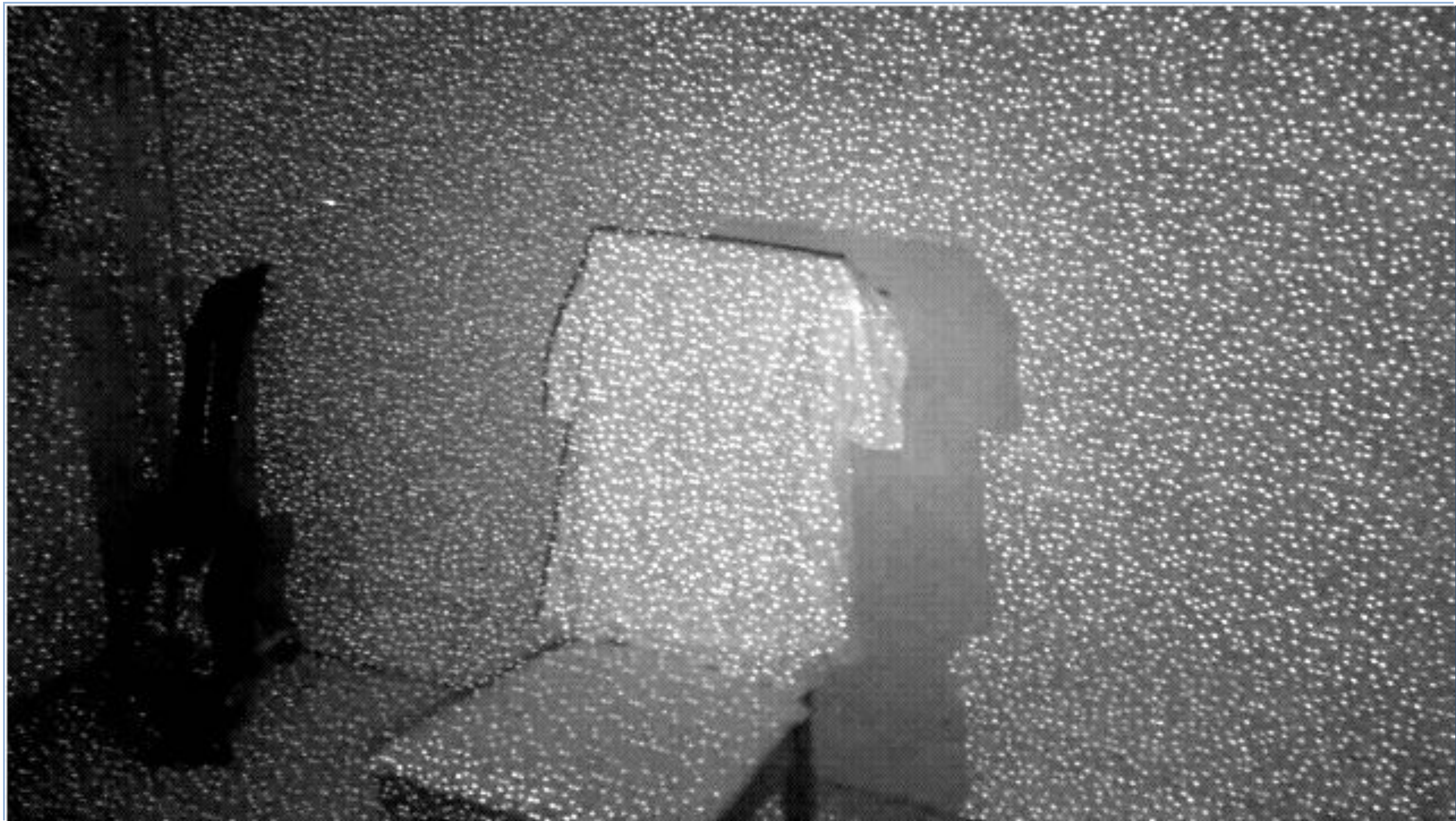
$w = 20$

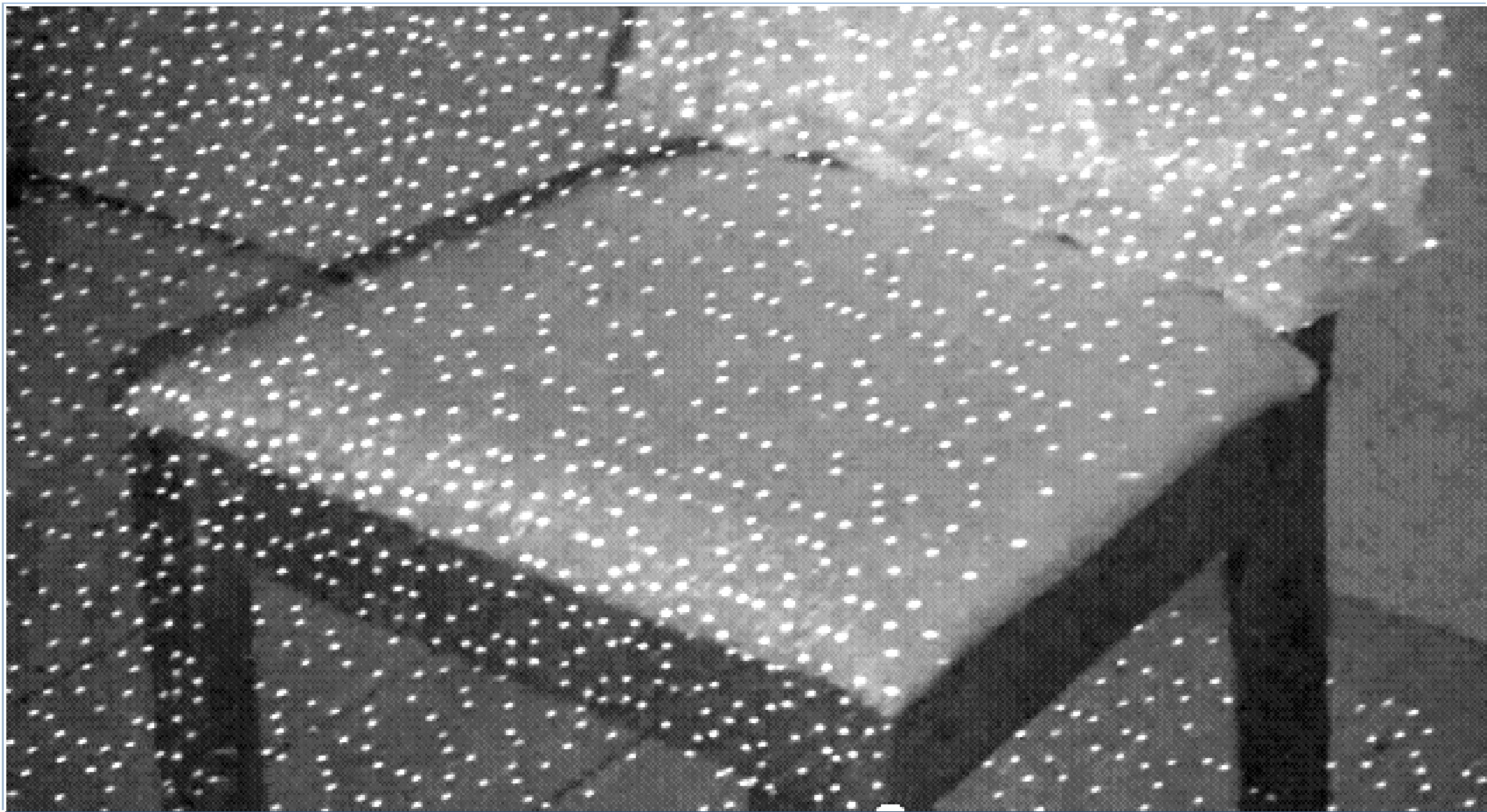


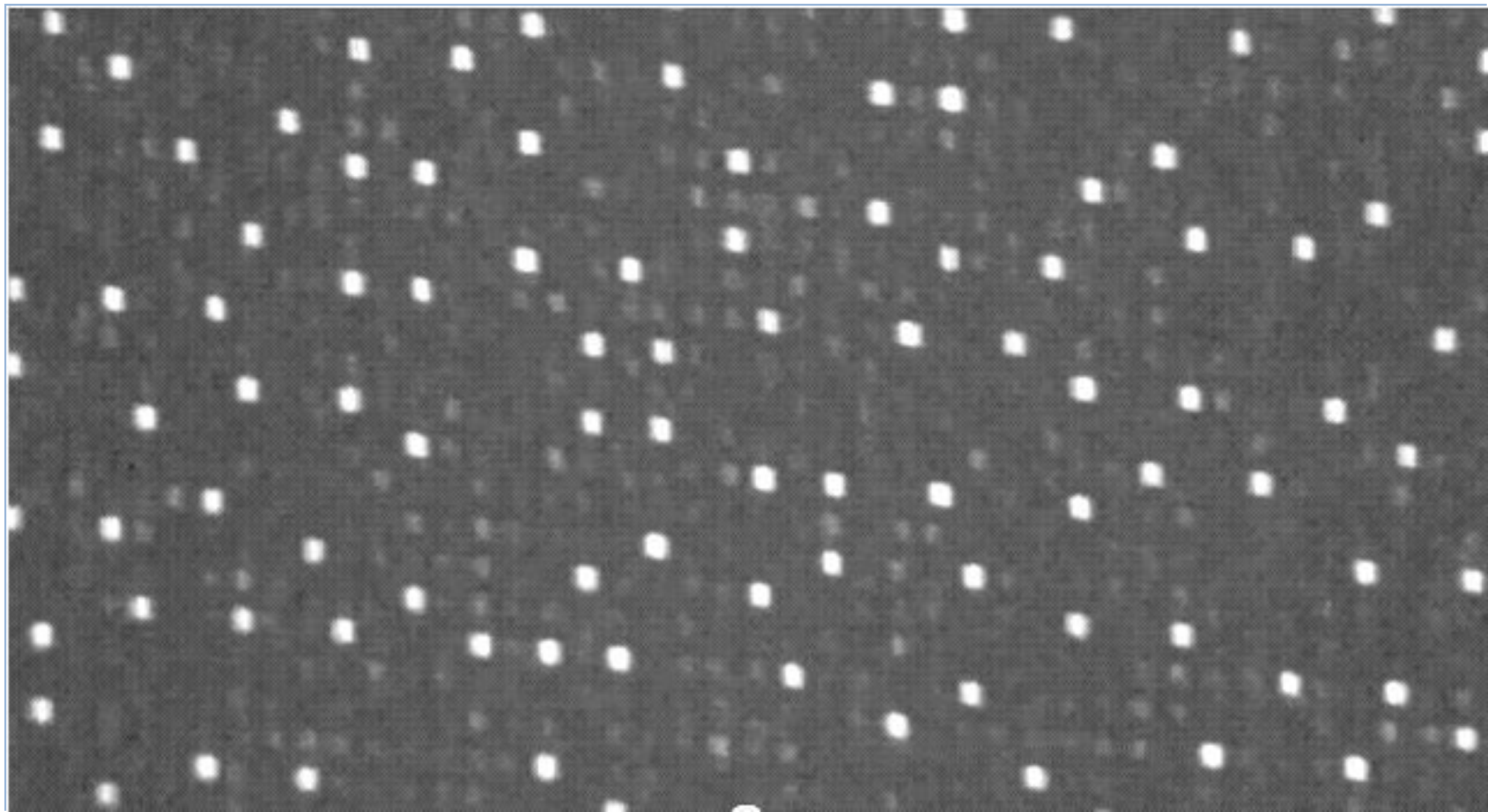
PrimeSense



Kinect







PrimeSense 3D Sensor

- Coded dot pattern in near IR - 830nm laser diode, visible to CMOS cameras. [Visible: 390-750nm. Thermal: 10,000nm]
- Pattern similar to “Blue Noise”, and is generated by a laser LED passing through a grating.
- Each 9×9 block in epipolar line gives a unique pattern.
- **Question:** Does the pattern change with distance?
- In every image location, search for a disparity up to 34 pixels along the epipolar line.
- Disparity becomes a lookup issue.
- Problem: Outdoors (Active light not strong enough)